ID: J-10-17 | Electronics & Electromagnets (Junior)

A Device to Limit Obesity

Krishna Bhatt | 7th Grade

My project has a goal of making children less prone to obesity by reducing the amount of calories children consume each day. I went about it by creating a child-friendly cookie dispenser which dispenses one cookie everyday. I added a bluetooth module and created an application so it could be controlled through a smartphone. The prototype overcame all of the success criteria with 100% accuracy. It also overcame my project constraints of budget, materials, and time. This design has scope to greatly reduce obesity rates. However it has to be mass produced and adopted by households across the U.S.



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The Problem and Inspiration



The CDC reports that “[o]besity is a serious health concern in the United States, affecting more than one in six children and putting their long-term health and quality of life at risk” (Lange, S.J., etc., 2021). Obese people are at a high risk of developing serious disorders such as coronary artery disease, strokes, type 2 diabetes, certain cancers, and it can interfere with breathing while asleep. The best way to prevent obesity in adulthood is to prevent obesity in childhood, as a study published by the National Library of Medicine reports that about half of school-age children and about a third of preschool children who are obese, become obese as adults.

My project was inspired by my younger sister who started eating more during the pandemic. I had researched further into this and had found that around 42.5% of people over twenty had obesity in the United States. The grabbed my attention, so I tried to design a solution to this pressing problem. I started working on this problem in January 2022.

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EXISTING SOLUTIONS and SIMILAR PRODUCTS

Dieting
Dieting means consuming less calories than you burn. It is not recommended for children.

Exercise
Exercise allows people to burn more calories than they intake. It is recommended for children.

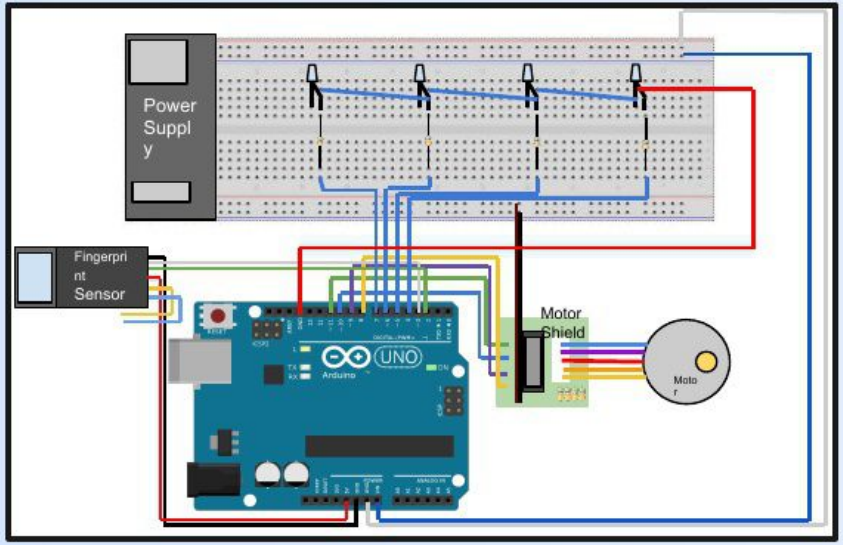
Sleeping
Research proves that those who sleep more are less likely to get obesity.

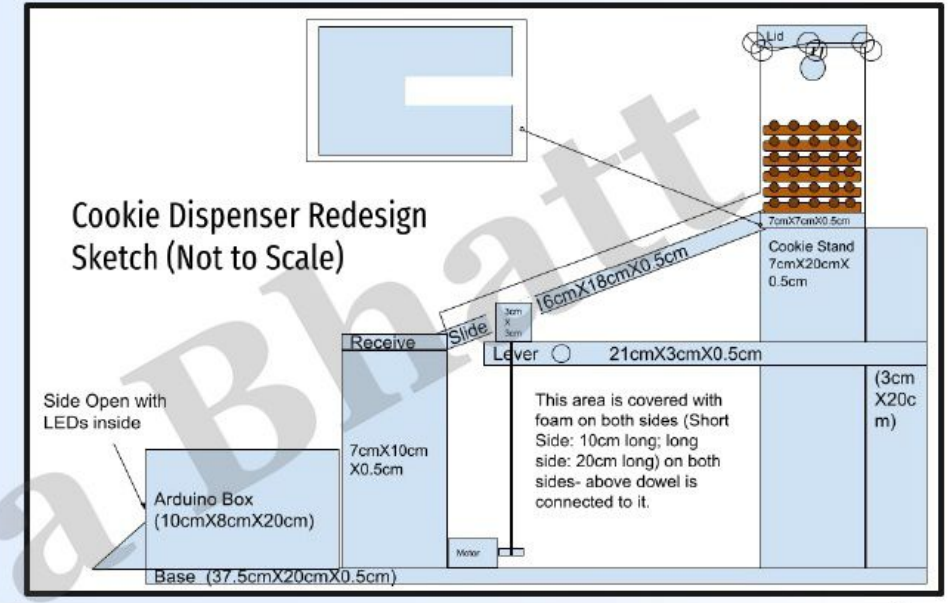
Gastric Bands
Allow less space for food in the stomach, limiting the amount of calories you intake.

Exercise Trampoline
Allow kids have aerobic exercise to reduce their weight, while they have fun.

GoNoodle®
An app that provides an interactive way for kids to get active.

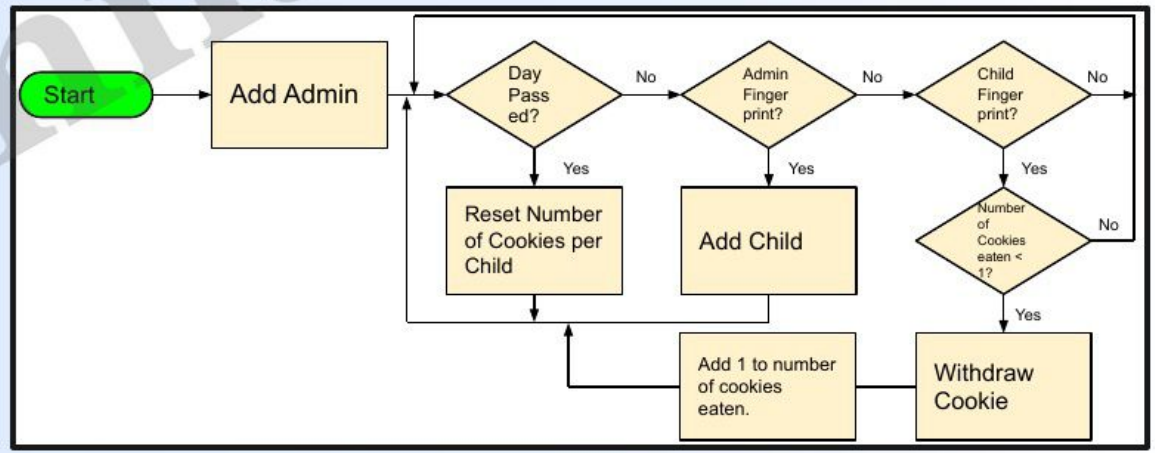
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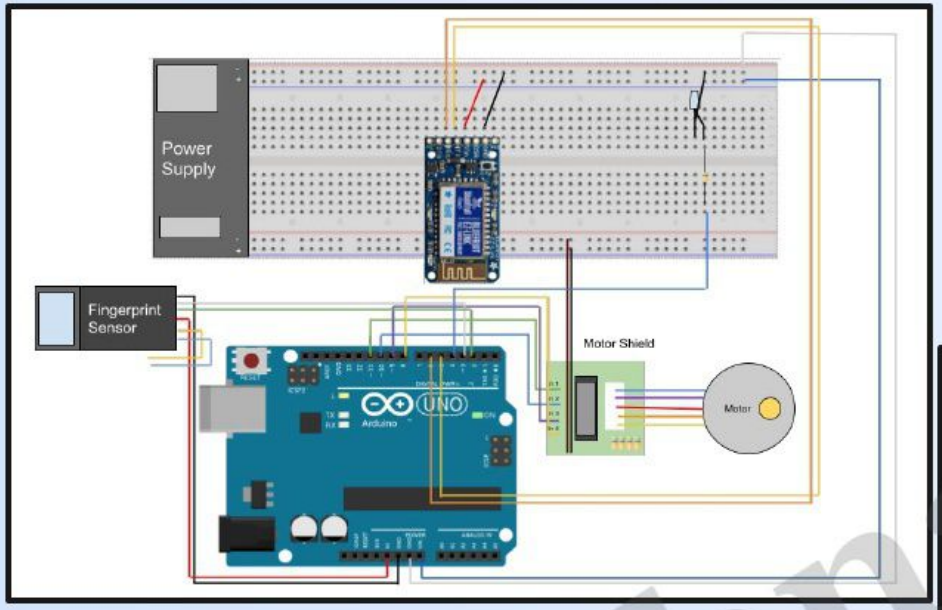


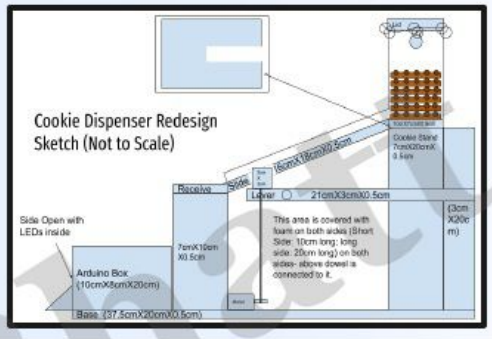
Design #1

A cookie dispenser that allows for a limited amount of cookies each day. It uses a lever to push a cookie out. Powered by an Arduino UNO, it uses a stepper motor and a fingerprint sensor. It uses LEDs to communicate with the user and the code follows the basic flowchart at the right.



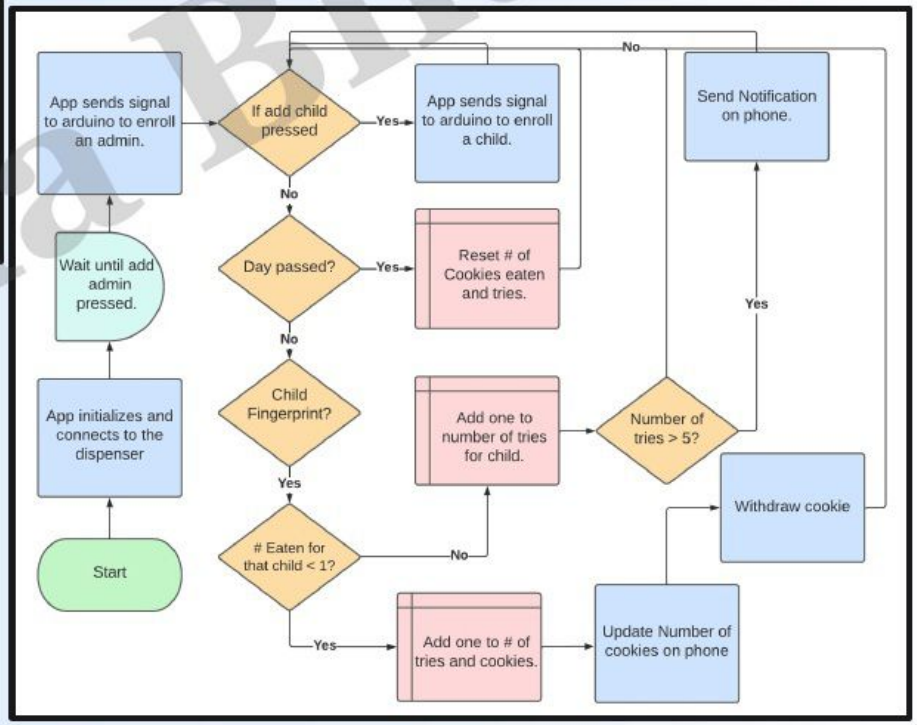
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Redesign

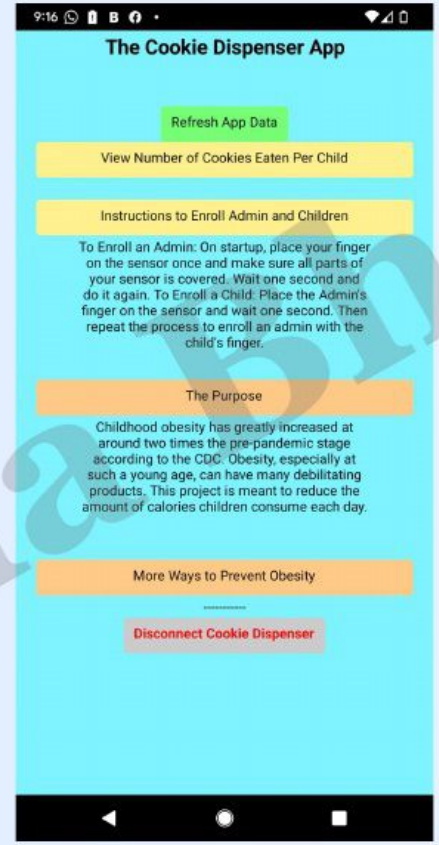
This is a redesign of the previous design and includes a bluetooth module and additions to the code. The structure is the same. An application was created to communicate with the Arduino through a smartphone. When a child is trying multiple times to gain a cookie, a notification is sent to the parent, and parents can check how many cookies a child has eaten.

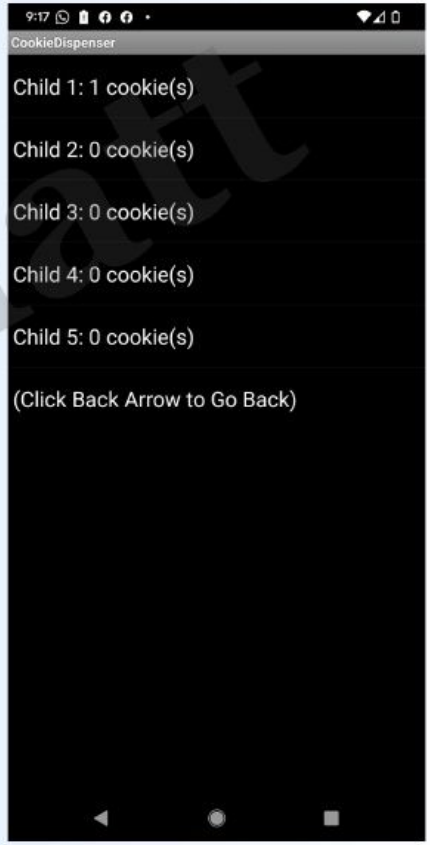


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Images of the App

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The Test Plans

The prototype should dispense only one cookie per day to each child.

Success Criteria #1

For 5 days, press fingerprints that are enrolled as children to the fingerprint scanner 5 times, and check how many cookies you receive per day.

Success Criteria #2

For 5 days, for each role, record the number of tries it takes to complete the default action.

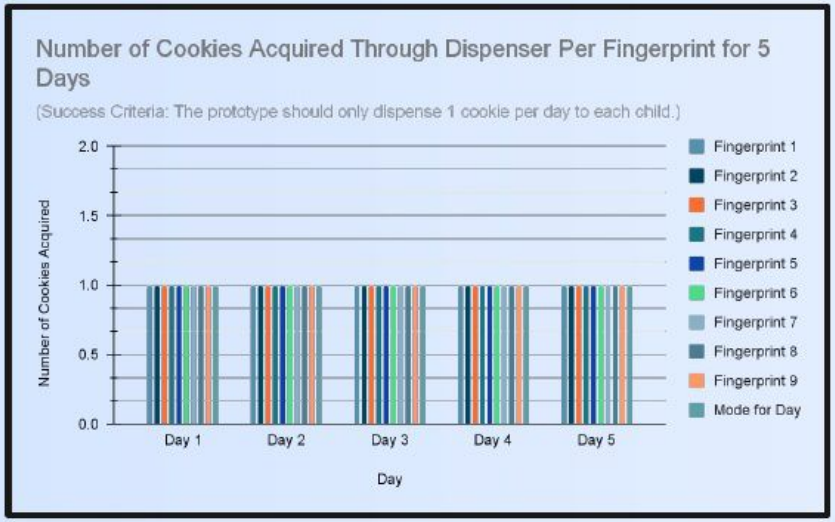
The prototype should successfully distinguish each fingerprint in the database.

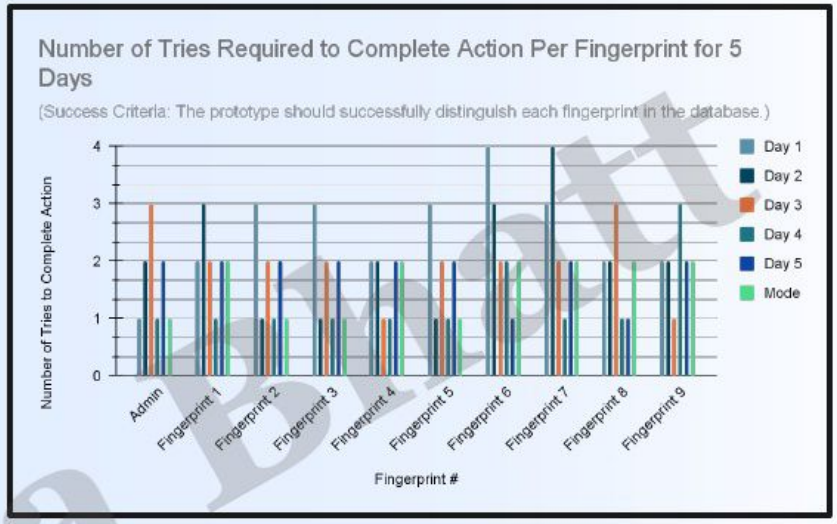
The prototype should give access to the cookie storage only to the admin.

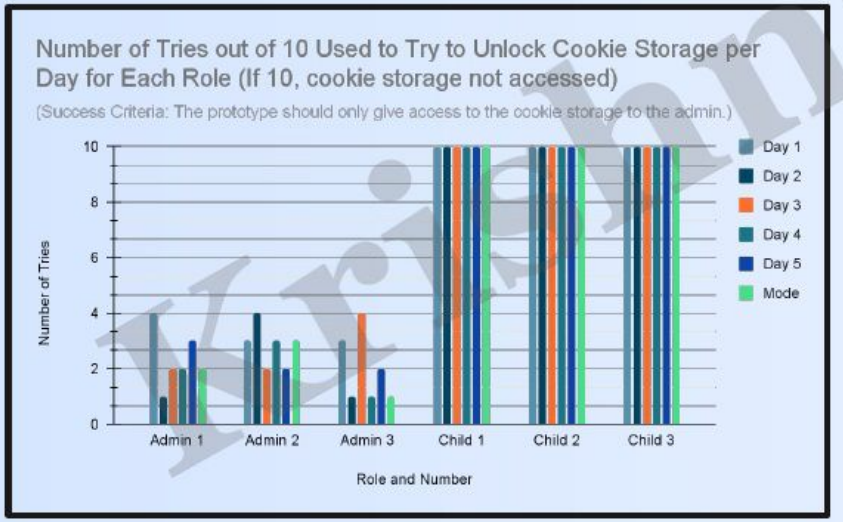
Success Criteria #3

Check how many times it takes for people who do not know the locker code to open the cookie storage and how many times for people who do for 5 days.

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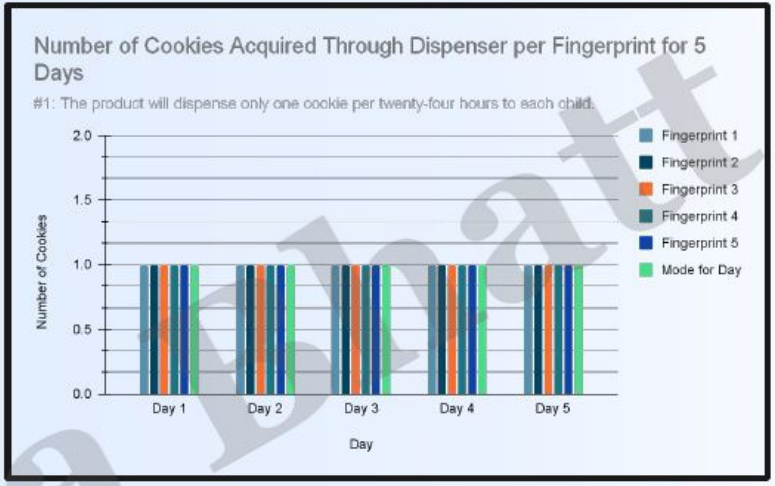
Design #1

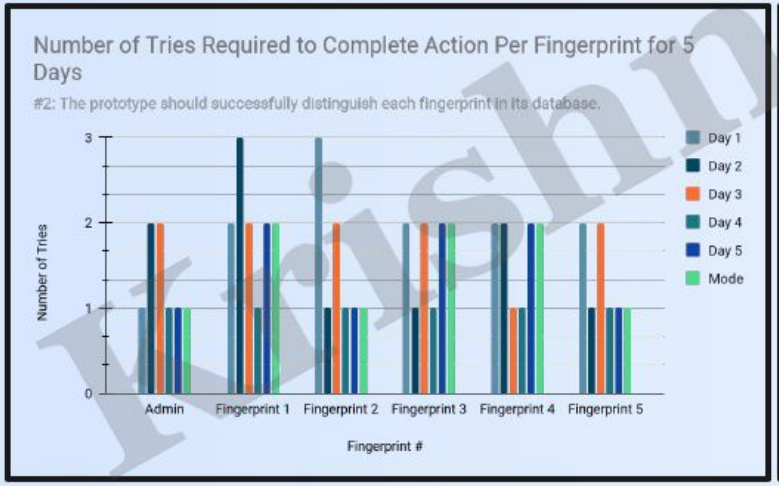
For the first design criteria, the prototype only dispensed one cookie per day to each person for every trial, and fulfilled this design criteria. For the second design criteria, the prototype's mode of the number of tries to complete its default action was 1. For the third success criteria, the prototype did not allow for any child to gain access to the cookie storage with 10 as the mode for the tables and graphs. All were met.

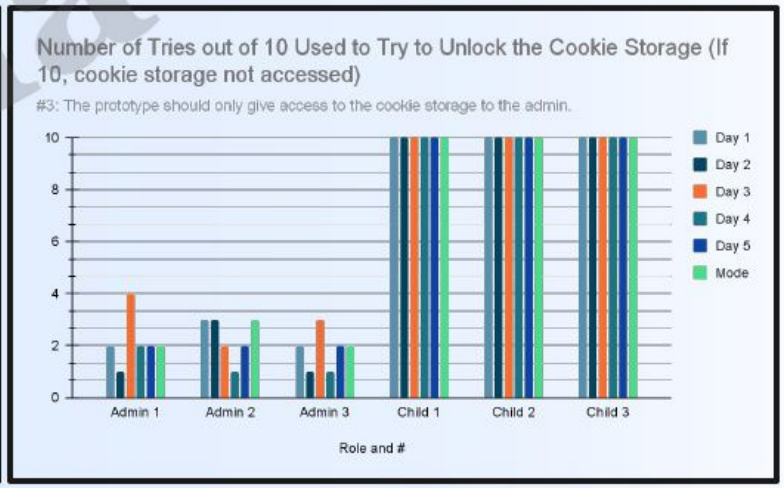
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Redesign

For the first design criteria, the prototype only dispensed one cookie per day to each person for every trial, and fulfilled this design criteria. For the second design criteria, the prototype's mode of the number of tries to complete its default action was 2. For the third success criteria, the prototype did not allow for any child to gain access to the cookie storage with 10 as the mode for the tables and graphs. All were met.







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Discussion

Design #1

- Passed all the design criteria
- Some scope to reduce obesity
- This is more efficient than other products as it is always in effect, and exercise programs only help when one takes an initiative.

Problems:

- Not very user friendly as it took multiple tries to complete some actions
- It takes 10 seconds to dispense
  - I don't have a faster motor to replace it

Redesign

- Passed all the design criteria
- Scope in reducing obesity in children.
- User-friendly design as it took less tries.
- Allows parents to have access to children's eating pattern which allows then to take more steps to improving the situation.

Problems:

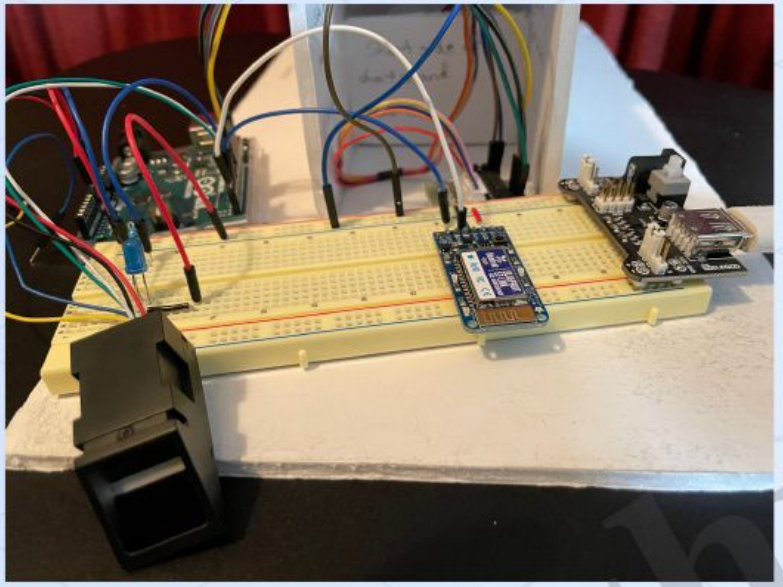
- Multi-threading could not be accomplished in MIT App Inventor
  - Refresh data button had to be made.

Future Improvements

- Constant refresh of app data.
- Using WiFi instead of bluetooth so the connection can be maintained at long distances.
- A compact design of the mechanism so it can fit in more places.
- A hard drive so data can be preserved after the plug is pulled out.

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Conclusion



Errors in Testing: Human error in testing the second and third success criteria

- I may have used the wrong finger
- Others may have completed the wrong motion while unlocking the lock

This could have butchered the data on how effective the dispenser was at detecting fingers and how user friendly it was.

This prototype can prevent children, and even adults, from having an excessive amount of carbohydrates everyday. Limiting the amount of calories taken each day and maintaining a healthy diet can be the first step to acquiring a healthy, normal weight.

The prototype that I have developed has a lot of scope to reduce obesity rates in the United States.

All the design constraints were overcome and many additions that I had not thought of previously came forward.

This model was effective as it only allowed kids to get one cookie per day, and it has multiple measures in place to prevent children from changing that..

The prototype successfully passed all success criterias and is considered a success. This project turned out better than I had expected.

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Supplemental Information

1. Lange, S.J., Kompaniyets, L., Freedman, D.S., Kraus, E.M., Porter, R., Blanck, H.M., and Goodman, A.B. Longitudinal Trends in Body Mass Index Before and During the COVID-19 Pandemic Among Persons Aged 2-19 Years. Morbidity and Mortality Weekly Reports (MMWR) - Centers for Disease Control and Prevention. September 17, 2021. December 24, 2021. https://www.cdc.gov/mmwr/volumes/70/wr/mm7037a3.htm?s_cid=mm7037a3_w.
2. Serdula, M.K., Ivery, D., Coates, R.J., Freedman, D.S., Williamson, D.F., and Byers, T. Do obese children become obese adults? A review of the literature. National Library of Medicine: National Center for Biotechnology Information: PubMed. March 22, 1993. December 24, 2021. https://pubmed.ncbi.nlm.nih.gov/8483856/.
3. Adafruit Fingerprint Sensor Library. Arduino. 2021. December 25, 2021. https://www.arduino.cc/reference/en/libraries/adafruit-fingerprint-sensor-library/.
4. Arduino Time library. Arduino Playground. 2021. December 25, 2021. https://playground.arduino.cc/Code/Time/.
5. Stepper Speed Control. Arduino. 2021. December 25, 2021. https://www.arduino.cc/en/Tutorial/LibraryExamples.
6. SoftwareSerial Library. Arduino. 2022. February 2, 2022. https://www.arduino.cc/en/Reference/softwareSerial.
7. 28BYJ-48 Stepper Motor with ULN2003 Driver and Arduino Tutorial. Makerguides. 2021. December 25, 2021. https://www.makerguides.com/28byj-48-stepper-motor-arduino-tutorial/.
8. DIYmall Optical Fingerprint Reader Sensor Module for Arduino Mega2560 UNO R3 51 AVR STM32 Red Light DC 3.8-7V FZ2904. DIYmall. 2016. December 27, 2021. http://www.diymalls.com/Optical-Fingerprint-Reader-Sensor.

Link to Hard Code (Google Drive):
Arduino Sketch:
https://drive.google.com/file/d/1DhmdaxXPz0O2AGXgS3JOQ6-I8c2tIVSi/view?usp=sharing
MIT App Inventor:
https://drive.google.com/file/d/1_F9DWtB8JWMxAy5Qe8wqkl3peMU-NgCh/view?usp=sharing

YouTube Demonstration (Redesign):
https://www.youtube.com/watch?v=uDxKYNIIAQ0

YouTube Demonstration (Original):
https://www.youtube.com/watch?v=1075X3V3PB0

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